Causal Failures in Network Systems

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***Abstract*—The failure of system components can cause serious damage to the performance of the system (e.g., link failures can disrupt flow in a network system). In traditional models, the failures of links are usually considered to be independent. However, link failures could be causal, wherein a link fails due to the failure of another non-adjacent link in the network. We call this kind of failure a *causal link failure*. In this paper, we formally define causal link failures in flow networks and investigate the impact of such failures on flow transmission. We formulate a discrete optimization problem to find the maximum number of causal link failures after which the flow network can deliver d units from source node s to terminal node t. We prove that the formulated problem is NP-hard and construct the corresponding optimization model. Furthermore, we present a greedy algorithm to approximately compute the solution to the problem. The experiments show the efficiency of the greedy algorithm compared to the optimization model.**

***Index Terms*—Causal failures, Flow networks, Link failures, Robustness**

Notation

$G = (N, A)$	A directed flow network
N	The node set
s, t	The source and terminal nodes
A	The link set
$A_j \subseteq A$	A subset of A
$a = (u, v)$	A link a from node u to v
A_j	A subset of A
C	The set of capacities for links
$c_{u,v}$	The capacity of a link (u, v)
$f_{u,v}$	The flow from u to v
$E : A_1 \Rightarrow A_2$	A causality from A_1 to A_2
$\mathcal{E}$	The set of all causalities
E_i	The i-th causality in $\mathcal{E}$
u, v-path	A path from u to v
d	The flow demand from s to t

I. Introduction

Network flow models have been applied to real-world applications, such as power transmission networks [1], supply chain networks [2], and transportation networks [3]. The failure of links affects the flow transmission in these networks and might trigger failures of other links or a situation in which no flow can be transmitted. More specifically, we consider the cases of the networks mentioned above as follows.

Power Systems: A power transmission system consists of generators, substations, and transmission towers, which are all modeled as nodes, and transmission lines, which are modeled as links connecting those nodes. Each transmission line consists of several physical lines, and each physical line supplies a specific capacity. Therefore, each transmission line has multiple capacities, depending on the number of physical lines operating. For example, if there are x functioning physical lines for a link, then we say that the capacity or state is x. Furthermore, if a transmission line fails, the load will be redistributed and other transmission lines might have to receive additional loads. It is possible that the extra load due to load redistribution exceeds the capacity of other transmission lines, and thus causes complete or partial failure of these lines if the load of the failed transmission line has a relatively large load. This is an example in which the failure of a link causes the failure of other non-adjacent links.

Supply Chain Systems: A standard supply chain network consists of suppliers, plants, warehouses, and retailers [4], and it is represented by nodes found within levels labeled 1, 2, 3, and 4, respectively. The transportation links are placed between levels l and $l + 1$, where $l = 1, 2, 3$. Each transportation link has its own flow capacity, representing the maximum number of products that could be shipped between levels. The demand originates from retailers, and the products are shipped through the network to satisfy that demand. If a link from a supplier fails due to a disruption (e.g., a natural disaster such as a flood or earthquake), the amount of each product delivered to the warehouses connected to the retailer will be reduced. As a result, the flow transmitted to the retailer could be smaller, or even 0. Let us consider that the flow of one link is 0 to be a link failure because of the initial link failure. We could observe from this example that a link failure in a supply chain network can trigger failures of other links, regardless of whether these links share common nodes.

Transportation Systems: A transportation network is a network in a geographical space, such as a highway or airline network. Such a network is generally modeled as a graph in which a node represents a location (e.g., a station, airport, or municipality), and a link is a connection between two locations. For example, in a road network, a link is a one-way road between two stations, and the weight of the link indicates the number of vehicles that a particular road can handle in a specific time period. Vehicles can bypass a link representing a closed road segment by finding alternate roads but which can cause severe traffic congestion. To avoid such congestion, a traffic authority can shut down some low-capacity roads, forcing vehicles to choose high-capacity roads. From the perspective of failures, if we consider the closed road segment to be a failed link, then causal link failures are a common phenomenon in road networks.

Causal Link Failures and the Contributions of this Paper: In the above networks, we see the failure of some links due to the failure of other links. We call this kind of failure a *causal link failure*. And we formalize the study of such causal link failures with the following contributions.

- We formally define causal link failures in a directed flow network.
- We formulate a problem to find the maximum number of causal link failures while preserving d units of flow to transmit from source node s to terminal node t.
- We prove the decision version of the formulated problem to be NP-complete and present a mixed linear programming model to find the exact solution to the problem.
- We design a greedy algorithm to approximate the solution to the optimization problem to deal with its computational complexity.

II. Background

A number of works have investigated situations similar to the causal failures explored here [5], [6], [7]. Such prior work includes network design considerations under correlated and causal failures [8], [9], [10]. Specifically related to causal failures as described in this paper, Zhang et al. [11] defined causal node failures in a single-layer network as the failure of a set of nodes due to the failure of another set of nodes. The authors formulated optimization problems and associated solution algorithms to find the size of the giant component and the number of connected components to address the robustness and vulnerability of a network given predefined causal failures. Failures are correlated not only within nodes but also between links. In the past 20 years, many researchers have been working on Shared Risk Link Groups, which are a feature allowing the user to establish a backup secondary label switched path or a fast-reroute label switched path which is disjoint from the path of the primary label switched path [12]. For the same Shared Risk Link Groups, the links represent the resources that share the same risk.

The flow network model is important in representing engineering applications. For example, the assignment problem, a fundamental problem in operations research, is widely applied in real life. Generally, it is defined as assigning k' agents to k jobs to minimize the corresponding cost. Hu et al. [13] constructed an integer programming model based on network flow and proposed a dynamic algorithm to solve the generalized assignment problem. In operator networks, network function virtualization technology helps manage network service, the requests of which are usually deployed as a service function chain. However, load imbalance has been an issue in such a network. Han et al. [14] proposed a method for implementing the service function chain using network flow theory to solve the load imbalance. For coupled electricity and gas networks, Mhanna et al. [15] formulated the multi-period optimal electricity and gas flow problem, which aims to find a dispatch such that the costs of power generators and nodes in the gas network are minimized and solve the problem by constructing algorithms based on network flow and linear programming.

Node/link failures affect the maximum flow and reliability between the source node s and the terminal node t in such flow networks. Specifically, the failure of one node triggers a malfunction of links emanating from the node. As a result, flow cannot be delivered through these malfunctioning links. Furthermore, the probability that d units of flow could be delivered from s to t is also lower because the failure of a node can reduce the number of minimal paths. Lin [5] considered a stochastic flow network where each link/node has several capacities and may fail, and proposed an algorithm to compute the probability that the maximum flow from s to t is at least a given integer d by finding all d-MPs and using the inclusion-exclusion principle. Similarly, Yeh [16] proposed an algorithm based on d-MC to compute multi-state two-terminal reliability based on the same network assumption. Both works show that node/link failures play a vital role in the reliability of a flow network.

These works do not mention the dependence of links in a network, regardless of whether the links share common nodes. We address this deficiency in the literature by exploring causal link failures and their impact on flow.

III. System Model and Problem Formulation

A. Model Description

We consider a directed flow network $G = (N, A, C)$, where N is the set of nodes, A is the set of links, and C is the set of link capacities. The network has $|N| = n$ nodes and $|A| = m$ links. Each link $a \in A$ can also be represented as an ordered pair of nodes $a = (u, v)$, where $u, v \in N$. Furthermore, the capacity of link $a = (u, v)$ is indicated by c_a or $c_{u,v}$, which is the maximum number of units of flow that can be transmitted through this link.

Fig. 1 depicts an example of a flow network which has six nodes $\{1, 2, 3, 4, 5, 6\}$ and nine links $\{a_1, a_2, a_3, a_4, a_5, a_6, a_7, a_8, a_9\}$, the capacities of which are $\{3, 2, 1, 2, 1, 3, 1, 1, 2\}$. For example, the capacity of link a_1 is 3, such that link a_1 can transmit at most 3 units of flow from node 1 to node 2. Links are correlated in this network. We consider the link dependence model as follows.

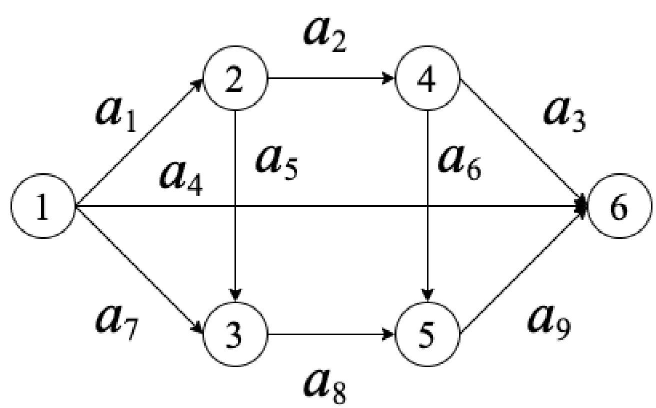

Fig. 1. An example of a flow network.

Definition 1. *Given a network $G = (N, A, C)$ and links $a_i, a_j \in A$, if the failure of link a_i triggers the failure of link a_j, we say that causal link failures, denoted $E : a_i \Rightarrow a_j$, are applied to G.*

Definition 2. *A causality $E : a_i \Rightarrow a_j$ is* applied *implies that when the link a_i is removed, a_j is removed.*

The removal, or failure, of a link is equivalent to the capacity of the link being set to 0. Furthermore, we generalize causal link failures to the case of link sets as follows.

Definition 3. *Given a network $G = (N, A, C)$, $A_1, A_2 \subseteq A$, if the failure of links in A_1 cause the failure of of* all *links in A_2, we call such failure as a* causal link failure *from A_1 to A_2. We refer to this as* failure of A_1 causes failure of A_2. *The causality is denoted as $E : A_1 \Rightarrow A_2$.*

Example 1. *Given the causality $E_1 : \{a_1\} \Rightarrow \{a_2\}$ in Fig. 1, if we apply E_1 to the network, then we remove links a_1 and a_2 or set the capacities of both links to 0 simultaneously. As a result, nodes 2 and 4 do not have incoming flow, and links a_3, a_5, and a_6 are unrelated to the flow transmission from node 1 to node 6.*

Note that causal link failures are an extension of the Shared Risk Links Group, a set of links that share a common resource, which affects all links in the set if the common resource fails [17]. Causal link failures can describe a variety of link dependencies from a network topology perspective.

B. Problem Formulation

For the flow network, the flow from source node s to terminal node t is affected once we apply causal link failures. Maximum flow is perhaps the most popular way to characterize flow in such networks, defined as the maximum amount of units that can be delivered from source node s to terminal node t. From the example in Fig. 1, the maximum flow from source node 1 to terminal node 6 is 5. Given causalities, $E_1 : \{a_1\} \Rightarrow \{a_2\}$, $E_2 : \{a_6\} \Rightarrow \{a_7\}$, and $E_3 : \{a_5\} \Rightarrow \{a_3\}$, applying them, respectively, reduces the maximum flow to 2, 4, and 4. As such, it is also interesting to explore the impact of causal failures on maximum flow, namely in finding the maximum subset of causalities that, when applied, enable the flow network to maintain some desired level of maximum flow.

Problem 1. *Given a flow network $G = (N, A, C)$, causalities $E_i : A_{1,i} \Rightarrow A_{2,i}, i = 1, 2, \cdots, h$, two nodes $s, t \in V$, and a constant d, with a maximum flow $f_{s,t}$, find a* **maximum** *subset of causalities whose application maintains $f_{s,t} \geq d$.*

Problem 1 evaluates the robustness of the network to maintain flow when faced with link failures. It uses the number of causalities of links as a metric instead of the number of links.

Intuitively, one could check each possible combination of all causalities and obtain the solution to Problem 1. However, the number of combinations to be checked is 2^h, where h is the number of causalities, indicating the hardness of Problem 1. In the following subsection, we prove the NP-hardness of Problem 1.

C. NP-Hardness Proof

We consider the decision version of the problem formulated, denoted as *VrEd*: Given graph $G = (N, A, C)$, causality set $\mathcal{E}$, source node s, and terminal node t, is there $\mathcal{E}' \subseteq \mathcal{E}$ with size at least r such that applying $\mathcal{E}'$ maintains $f_{s,t} \geq d$?

To prove *VrEd* is NP-complete, we refer to the decision version of the maximum 2-satisfiability (MAX-2SAT) problem: Given a Conjunctive Normal Form with p clauses $c_1, c_2, \ldots, c_p$, is there a true/false assignment of all literals such that there are at least r clauses that are satisfied? MAX-2SAT is NP-complete [18]. Then we have the following.

Theorem 1. *VrEd is NP-complete.*

Proof. First, it is straightforward to see that *VrEd* is in NP because we could verify a certificate of *VrEd* in polynomial time by using a known algorithm to find the maximum flow.

Then we perform a polynomial-time reduction from an instance I of MAX-2SAT to an instance I' of *VrEd* as follows:

- We identify source node s and terminal node t. Suppose that there are l literals $x_1, x_2, \ldots, x_l$ and/or their negatives $\bar{x}_1, \bar{x}_2, \ldots, \bar{x}_l$. For each literal x_i, we have nodes X_i, their negatives $\bar{X}_i$, and a dummy node D_i. In addition, we have links (s, X_i), $(s, \bar{X}_i)$, (X_i, D_i), $(\bar{X}_i, D_i)$ and (D_i, t). The capacities of all the links here are set to p.
- For each clause $c_j = x_a \vee x_b, j = 1, 2, \ldots, p$, we construct nodes x_a, x_b, c_j and links (s, x_a), (s, x_b), (x_a, c_j), (x_b, c_j), (c_j, t). The capacities of all the links here are set to p. In particular, if x_a appears in more than one clause, we do not construct duplicate nodes for x_a, that is, only one x_a is enough in the graph.
- If x_a is in clause $c_j, j = 1, 2, \ldots, p$, causality is defined as $E_a : \{(s, x_a)\} \Rightarrow \{(x_a, c_j), (s, X_a)\}$.
- Let $d = pl + p^2 - p + 1$ and E_a be applied if x_a is assigned True in clause c_j.

Suppose I is satisfied, then at least r clauses are satisfied. Then we choose one true literal from each satisfied clause, and consequently, at least r causalities are applied. The flow from s to t is $pl+p$, which meets the requirement that at least d units are transmitted from s to t.

Suppose I is not satisfied; then, at most, $r-1$ clauses are satisfied. Further, we have the following.

Claim: It is not possible that causalities r are applied to the network.

Suppose there are r applied causalities. If each comes from one clause, then we have r clauses that are satisfied according to the above construction. Therefore, two of them come from the same clause c_j, leading to the situation that no flow from s to t will flow through node c_j. As a result, the flow would be at most $p(l-1)+p^2 = pl+p^2-p < d$. Alternatively, two of the applied causalities come from x_i and $\bar{x}_i$, then the flow from s to t through node D_i is 0. Consequently, we have $f_{s,t} \leq pl + p(p-1) = pl + p^2 - p < d$. Therefore, the claim holds and the theorem is proved. □

For another explanation of the proof of theorem 1, we provide an example in Fig. 2 where the instance of MAX-2SAT is $(x_1 \vee x_2) \wedge (\bar{x}_1 \vee x_2) \wedge (\bar{x}_1 \vee \bar{x}_2)$. Consequently, six causalities are constructed in the graph, and, finally, two causalities are applied to the network according to the assignment of literals.

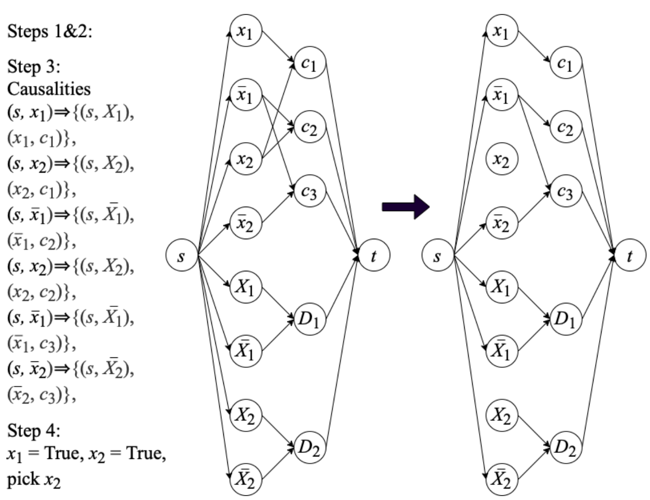

Fig. 2. Explanation of proof for Theorem 1

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IV. Methods

A. Optimization Model

We provide a mixed-integer linear programming (MILP) model to solve *VrEd*. In this model, an additional link is added from node t to s, (t,s), with infinite capacity, and z_E is a binary variable that indicates whether causality E is applied: $z_E = 1$ indicates that causality E is applied, and $z_E = 0$ indicates that causality E is not applied.

$$
\begin{aligned}
maximize \quad & \sum_{E\in\mathcal{E}} z_E && (1)\\
s.t.\ f_{t,s} \quad \geq \quad & d && (2)\\
\sum_{u\in N} f_{v,u} \quad = \quad & \sum_{u\in N} f_{u,v}, v \in N && (3)\\
f_{u,v} \quad \leq \quad & c_{u,v}, (u,v) \in A && (4)\\
f_{u,v} \quad \geq \quad & 0, (u,v) \in A && (5)\\
f_{u,v} \quad \leq \quad & c_{u,v}(1-z_E), E \in \mathcal{E}, (u,v) \in A(E) && (6)\\
z_E \quad \in \quad & \{0,1\}, E \in \mathcal{E} && (7)
\end{aligned}
$$

This formulation aims to find the maximum number of causalities with Eq. (1) while preserving the flow transmitted from s to t is at least d, governed by Eq. (2). We assume that there is no learning effect or deterioration effect in the network, so for each node, the sum of all incoming flows equals the sum of all outgoing flows, as dictated by the flow conservation law in Eq. (3). The flow passing through each link does not exceed the capacity of the link and must be nonnegative, ensured by Eqs. (4) and (5), respectively. For the links of causality E, Eq. (6) requires that the flow must be 0 if causality E is applied, according to Eq. (7).

B. Greedy Algorithm

To find the solution more efficiently, we propose an approximation algorithm considering causalities. As we know, applying one causality might reduce the maximum flow from the source to the terminal. If the maximum flow drops sharply after removing links from one causality, then this causality should not be applied to the network first. Therefore, for causality $E \in \mathcal{E}$, we consider the maximum flow in the graph $G-E$, denoted as $\theta(E)$, and classify the causalities based on $\theta(E)$ in descending order. Moreover, we use a stack to store the applied causalities while checking whether the maximum flow is at least d. Details of the algorithm are listed below.

Algorithm 1 Approximation for Max Causality (AMC)

Input: graph $G = (N, A, C)$, causality set $\mathcal{E}$, flow demand d, source s, terminal t;
Output: approximated the maximum number of applied causalities.

Sort all causalities based on θ values in ascending order, and let $\theta(E_1) \geq \theta(E_2) \geq \cdots \geq \theta(E_h)$;
MC $= \emptyset$; *//initialization*
for i in $1:h$ **do**
  if the maximum flow of $G - E_i$ is at least d **then**
    $MC \leftarrow MC + E_i$; *//updating MC*
    $G \leftarrow G - E_i$;
  else
    Continue;
  end if
end for
return MC.

Computational Complexity: Algorithm 1 only needs to compute the maximum flow for $2h$ times in the worst case. Hence, the total computational complexity is $O(hnm^2)$.

Approximation Ratio: After sorting, the first causality can be applied to the network, but applying each of the remaining causalities would reduce the maximum flow to less than d, while applying all other causalities is the solution. Therefore, this worst-case scenario indicates that the tightest approximation ratio should be $h-1$.

V. Results

In this section, multiple experiments are presented to compare the performance of the proposed algorithms and the optimization model. All algorithms, coded in Python, are implemented on a desktop with a 2.6 GHz 6-Core Intel Core i5 with a RAM of 16GB. Specifically, the optimization model utilizes the Gurobi library [19] and the NetworkX library [20] to help code the greedy algorithm and the genetic algorithm.

A. Comparison of Performance of Algorithms

In this subsection, we use a power network, denoted power2000, with 1,723 nodes and 2,394 links from Rossi and Ahmed[21] with the following settings.

- We assign the capacity of each link by randomly sampling an integer from 1 to c_{max}. In this experiment, $c_{max} = 20$.
- For each causality $E : E_1 \Rightarrow E_2$, we randomly generate n'_1 and n'_2 links for E_1 and E_2, respectively, where $n'_1, n'_2 \in [1, n_{max}]$. In this experiment, $n_{max} = 30, h = 40$. In addition, to avoid cascading failures, we ensure that all causalities are disjoint.
- When running the optimization model, we convert this undirected network to a directed network by replacing each link uv with two links (u, v) and (v, u), the capacities of which are the same.

We choose different pairs of s, t, compute the number of causalities, and list the results in Table I.

From Table I, we observe the efficacy of the greedy algorithm. The greedy algorithm achieves 88% of the solution or better (and achieves the solution in around half of the experiments). For example, the optimization model and the greedy algorithm obtained 36 causalities for $s = 40, t = 310, d = 1$. Similarly, for $s = 20$, $t = 1000$, and $d = 9$, the number of causalities applied as obtained by the greedy algorithm is only 1 less than that obtained by the optimization model. However, in some cases, the greedy algorithm spent more time getting the results than the optimization model. For example, it took the optimization model 6.17 seconds to get the solution for $s = 500, t = 1500, d = 16$, while the greedy algorithm consumed 9.68 seconds. Similarly, for $s = 20, t = 1000, d = 9$, the greedy algorithm spent about 1 second more than the optimization model. This indicates that the greedy algorithm does not necessarily provide an advantage relative to the optimization model in terms of the running time for power2000.

From the perspective of the relationship between the number of links and the number of causalities, we could observe the advantage of causalities from the experiments. Consider a network with m links where we are looking for the maximum number of failed links such that there exists flow d from s to t. This problem is very hard, requiring $O(2^m)$ time in the worst case because one needs to check each possible combination of all links. However, based on causalities, we know that the expected number of links in one causality is $n_{max}+1$ and the expected number of causalities is $\frac{m}{n_{max}+1}$ if all links belong to a specific causality. Consequently, we need to check the possible combination of causalities, which takes much less time than $O(2^m)$.

VI. Concluding Remarks

This paper defines the causal link failures and their impact on the flow network. We formulate an optimization problem to investigate the robustness of the flow network and design a greedy algorithm to search for the solution exactly and heuristically. The experiments show the efficiency and efficacy of the greedy algorithm for large-scale networks.

In future work, we will investigate probabilistic causal failures. More specifically, we plan to define the probability of applying one causality based on the likelihood that each link is functioning. Then the reliability of a flow network facing causal failures will be defined. Moreover, we will also look at cascading causal link failures in flow networks.

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TABLE I
PERFORMANCE COMPARISON OF OPTIMIZATION MODEL AND GREEDY ALGORITHM FOR POWER2000.

s, t	distance	max-flow	d	OPT		AMC		
				mc_{opt}	$t_{opt}(s)$	mc_{amc}	t_{amc}(s)	mc_{amc}/mc_{opt}
$\{40, 310\}$	15	5	1	36	6.10	36	4.75	1.00
			2	36	5.88	35	4.72	0.97
			3	35	6.09	35	5.17	1.00
			4	34	5.95	34	4.96	1.00
			5	33	6.09	33	5.10	1.00
$\{20, 1000\}$	14	11	1	36	5.81	36	6.80	1.00
			3	36	5.97	36	6.72	1.00
			5	36	5.85	36	6.64	1.00
			7	34	5.82	31	6.53	0.91
			9	32	5.91	30	7.12	0.94
$\{500, 1500\}$	11	18	1	38	5.81	37	8.70	0.97
			4	38	5.84	38	8.60	1.00
			7	37	6.04	35	8.79	0.94
			10	35	5.97	31	9.94	0.88
			16	29	6.17	28	9.68	0.96

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